RanSMAP 2024 – EDA Report

ATA-read behavioral features: ransomware vs benign

1. Executive summary

- The dataset is large (160.26 GB) and well balanced: 6 ransomware families and 6 benign applications, with near-equal file counts per class.
- ATA-read files are clean: no missing values in any of the 1,799 files, and only a handful of duplicate rows in benign files.
- The clearest class signal is volume: ransomware files average about 485K rows versus about 285K for benign (roughly 1.7x).
- Ransomware files contain far more duplicate rows (mean 3.6 per file, max 111) than benign files (mean 0.04, max 2).
- Only columns 0–3 carry information; columns 4 and 5 are constant (-1 and 0) in both sample files and can likely be dropped.

2. Dataset overview

The notebook measured 160.26 GB on disk. The dataset documentation splits this as follows:

Subset	Size	Purpose
Original (training)	111.8 GB	Ransomware and benign recordings on two CPUs: 63.23 GB (Celeron gen6) and 49.60 GB (i3 gen12)
Extra	28.57 GB	Additional records
Mix	10.58 GB	Ransomware running alongside benign apps, to test simultaneous detection
Variants	9.28 GB	7 Conti variants, to test detection of unseen variants

Ransomware: Conti, Darkside, LockBit, REvil, Ryuk, WannaCry. **Benign:** AESCrypt, Firefox, Idle, Office, SDelete, Zip.

Files were grouped by name into five behavioral data types:

Data type	Benign files	Ransomware files
ATA-read	900	899
ATA-write	900	900
Memory-read	1,800	1,800
Memory-write	900	900
Memory-execution	900	900

Classes are almost perfectly balanced. ATA-read has one fewer ransomware file (899 vs 900), and Memory-read has twice as many files as the other types; both are worth verifying.

3. Data quality (all 1,799 ATA-read files)

Every file has 6 columns and no header (read with `header=None`).

Metric	Benign (n=900)	Ransomware (n=899)
Columns	6	6
Missing values	0	0
Rows per file, mean	285,255	485,166
Rows per file, min	676	1,100
Rows per file, max	1,566,595	1,633,102
Duplicate rows per file, mean	0.036	3.60
Duplicate rows per file, max	2	111

Findings

1. **Completeness:** no missing values anywhere, so no imputation is needed.
2. **Volume gap:** ransomware recordings are about 70% longer on average. Ranges overlap heavily (similar maxima), so row count alone is not a reliable classifier.
3. **Duplicates:** benign files are essentially duplicate-free; ransomware files occasionally contain many repeated rows (up to 111). Repeats in sequential disk reads are plausible behavior, so keep them for modeling by default.
4. **Wide spread:** file sizes range from under a thousand to over 1.5 million rows in both classes, so use windowing or fixed-length sampling rather than whole-file inputs.

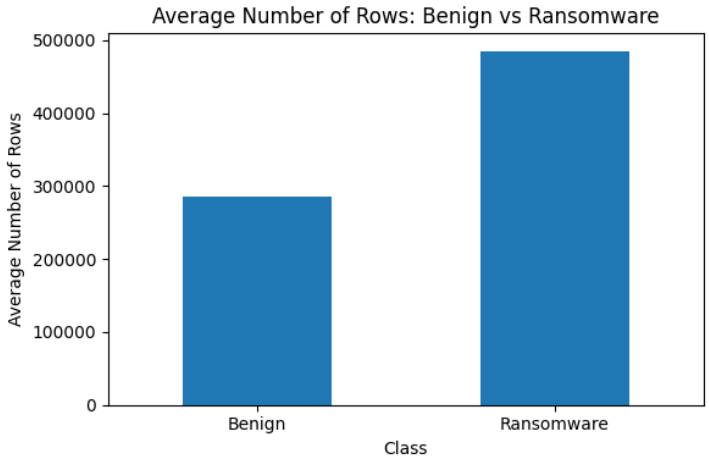


Figure 1. Average number of rows per file, benign vs ransomware.

4. Sample file deep dive

One randomly chosen file per class was inspected. These numbers illustrate typical structure and should not be read as class-wide statistics.

Property	Benign sample	Ransomware sample
Shape	102,985 x 6	180,028 x 6
Data types	All int64	All int64
Missing / duplicates	0 / 0	0 / 0
Col 3 mean (std)	3,958 (583)	3,884 (702)
Col 2 mean (std)	6.93e7 (4.08e7)	4.42e7 (3.45e7)
Col 2 max	1.66e8	1.66e8
Col 1 range	about 4e4 to 1e9	about 5e2 to 1e9

Column interpretation

The columns are unlabeled in the notebook. A plausible reading, inferred from the values and not confirmed, is:

- **Column 0:** Unix timestamp in seconds; both recordings span roughly 200 seconds.
- **Column 1:** sub-second timestamp in nanoseconds (0 to 1e9), roughly uniform with mild peaks near zero.
- **Column 2:** disk position or address; early rows grow in steps of 8, which looks like sequential sector addressing.
- **Column 3:** request size in bytes, dominated by 4,096 with a small tail down to 512.
- **Columns 4 and 5:** constant at -1 and 0 (zero variance).

Distribution observations

- **Block size (col 3):** overwhelmingly 4,096 bytes in both classes; the tail of smaller requests is slightly heavier in the ransomware sample (std 702 vs 583).
- **Address (col 2):** the benign sample is multimodal across the full range (median about 6.0e7). The ransomware sample concentrates below about 6.5e7 (median about 3.9e7), shows a gap between roughly 6.5e7 and 9.5e7, then a smaller cluster at 1.0e8–1.6e8. This hints at reads from a narrower disk region, based on one file per class.
- **Timestamps:** the ransomware sample shows a sharp early spike in col 0, consistent with a burst of reads at process start; the benign sample peaks mid-recording.
- **Constant columns:** columns 4 and 5 carry no discriminative information.

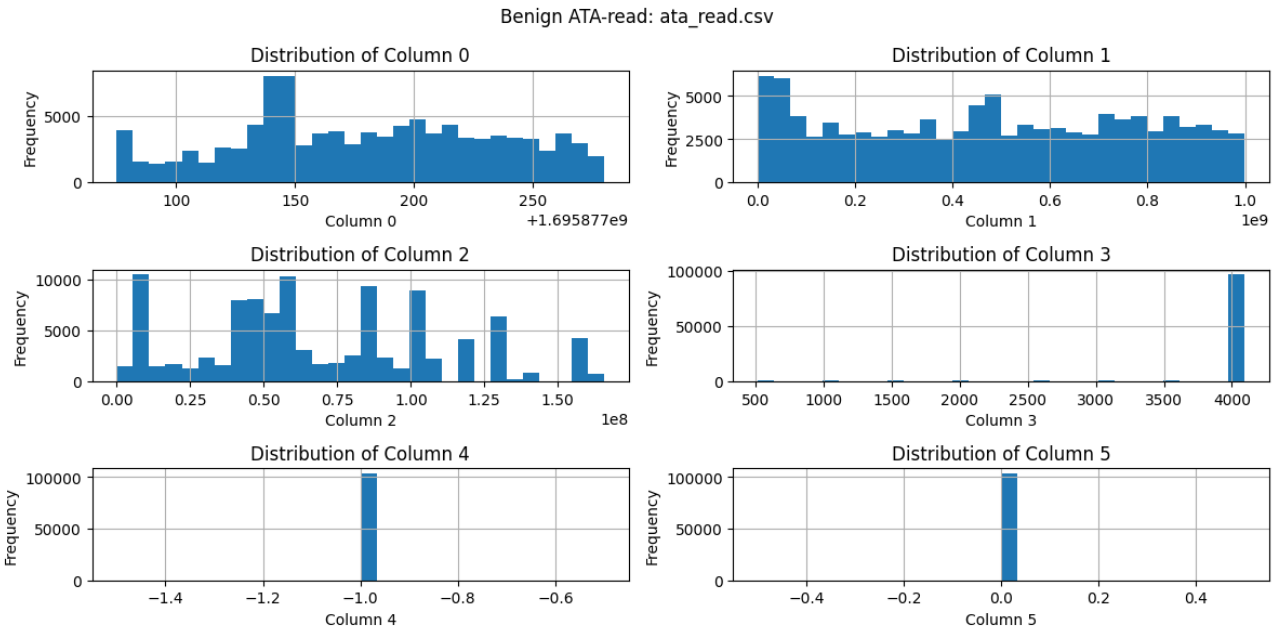


Figure 2. Column distributions, benign ATA-read sample.

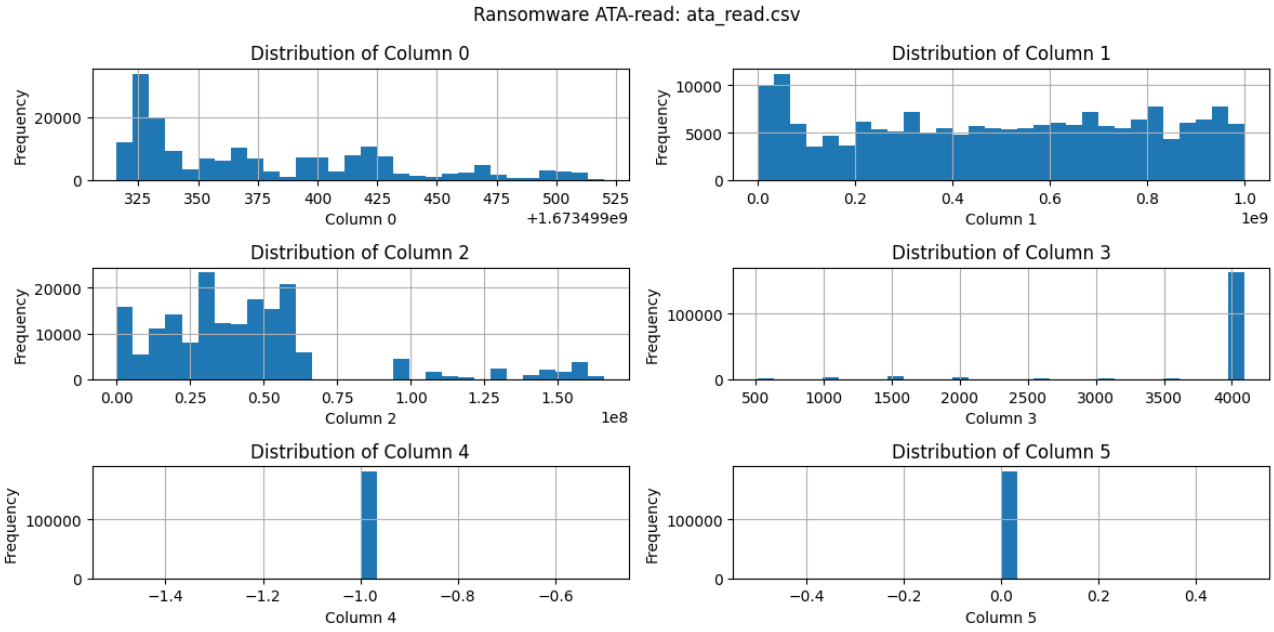


Figure 3. Column distributions, ransomware ATA-read sample.